



# Introduction

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# Computer Graphics

- Computer Graphics is the process of rendering static scene or animation on computer screen in an efficient way.
- Simply, computer graphics is computer generated image or scene.
- Scene might include text, color, pictures.
- The scene can be from imaginary or real world.
- The objects in the scene might have different geometric shapes and colors.
- Each object is rendered on the screen with different style and size and color.

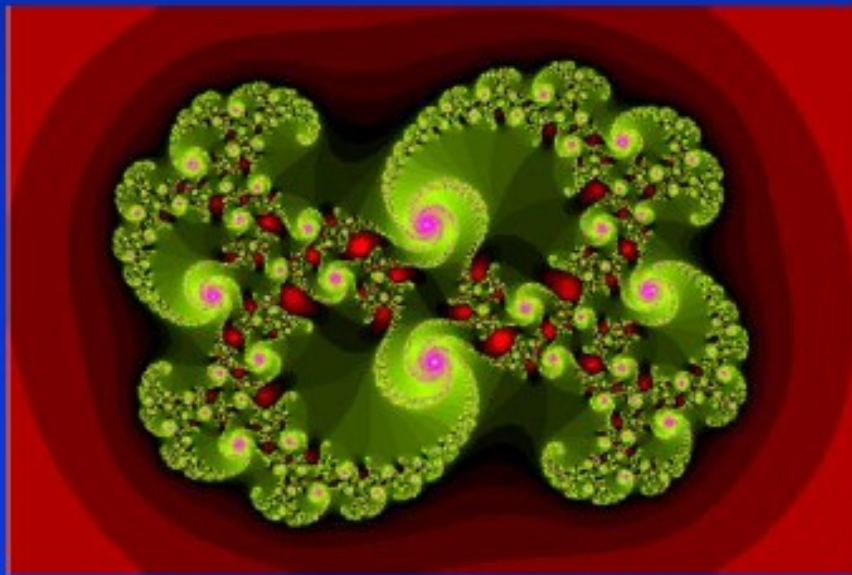
# What is Computer Graphics?

- The creation of, manipulation of, analysis of, and interaction with pictorial representations of objects and data using computers.
  - Dictionary of Computing
- A picture is worth a thousand words.

## - Chinese Proverb

1000 words (or just 94 words), many letters though...

It looks like a swirl. There are smaller swirls at the edges. It has different shades of red at the outside, and is mostly green at the inside. The smaller swirls have purple highlights. The green has also different shades. Each small swirl is composed of even smaller ones. The swirls go clockwise. Inside the object, there are also red highlights. Those have different shades of red also. The green shades vary in a fan, while the purple ones are more uni-color. The green shades get darker towards the outside of the fan ...



# Why Computer Graphics?

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- About 50% of the brain neurons are associated with vision
- Dominant form of computer output
- Enable scientists (also engineers, physicians, and general users) to observe their simulation and computation
- Enable them to describe, explore, and summarize their datasets (models) and gain insights
- Enrich the discovery process and facilitate new inventions

# Why Computer Graphics?

- Applications (In essence, computer graphics is application-driven)
  - Entertainment: Movies, Video games
  - Graphical user interface (GUI)
  - Computer aided design and manufacturing (CAD/CAM)
  - Engineering analysis and business
  - Medical applications
  - Computer Art
  - Engineering Analysis
  - Scientific visualization / simulation
  - Virtual Reality

# Movies

- If you can image it, it can be done with computer graphics!
- More than one billion dollars on special effects.
- No end in sight for this trend!



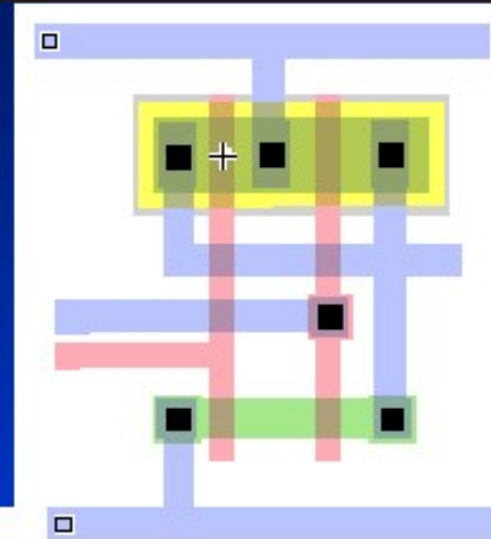
# Video Games

- Important driving force
- Focus on interactivity
- Try to avoid computation and use various tricks



# Computer-Aided Design

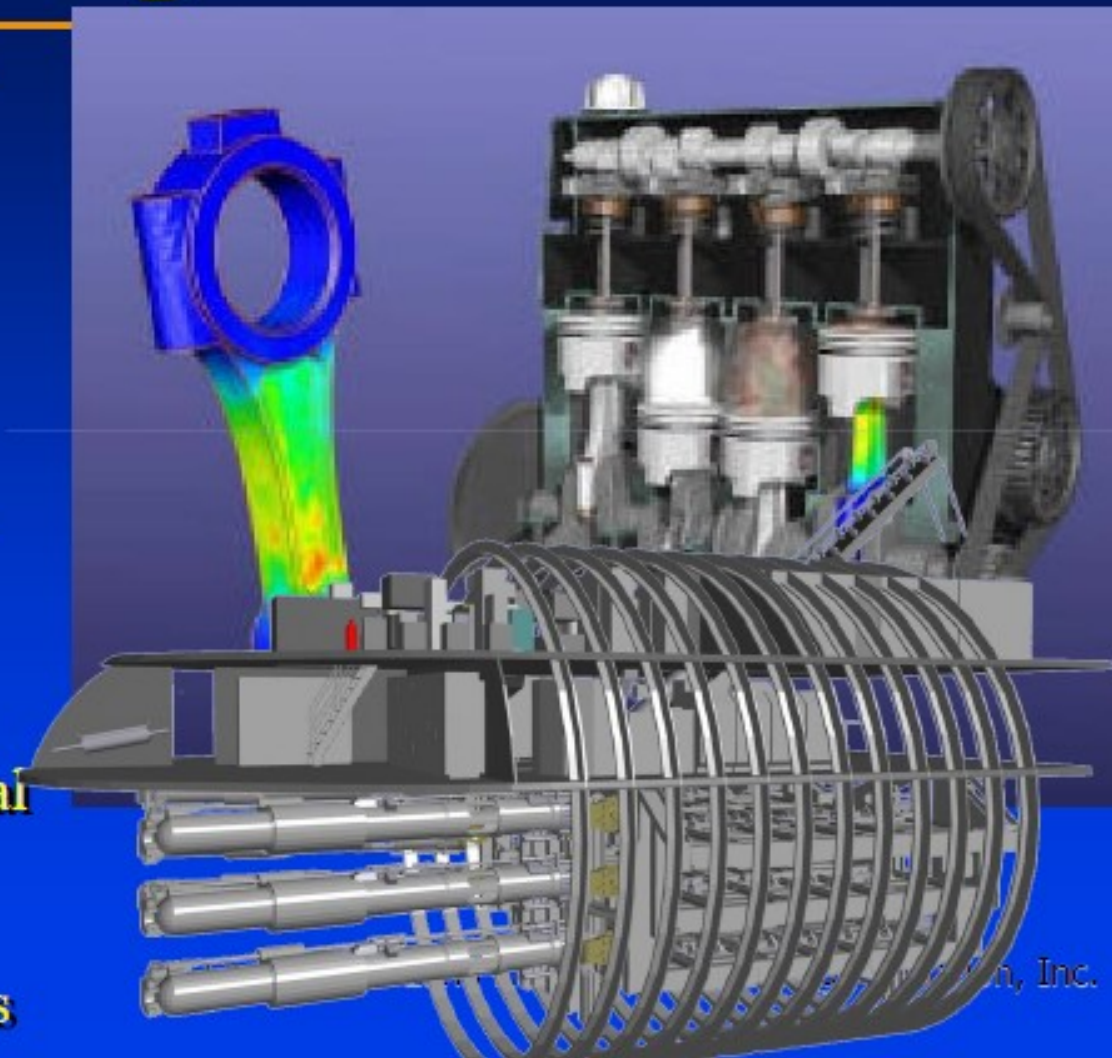
- Significant impact on the design process
- Mechanical, electronic design
  - entirely on computer
- Architectural and product design
  - Migrate to the computer



UGS: towards virtual manufacturing

# Engineering Design

- Engineering & Architecture Software
- Buildings, aircraft, automobile, computers, appliances, etc.
- Interactive design (mesh editing, wire-frame display, etc.)
- Standard shape database
- Design of structural component through numerical simulation of the physical operating environment
- Testing: real-time animations



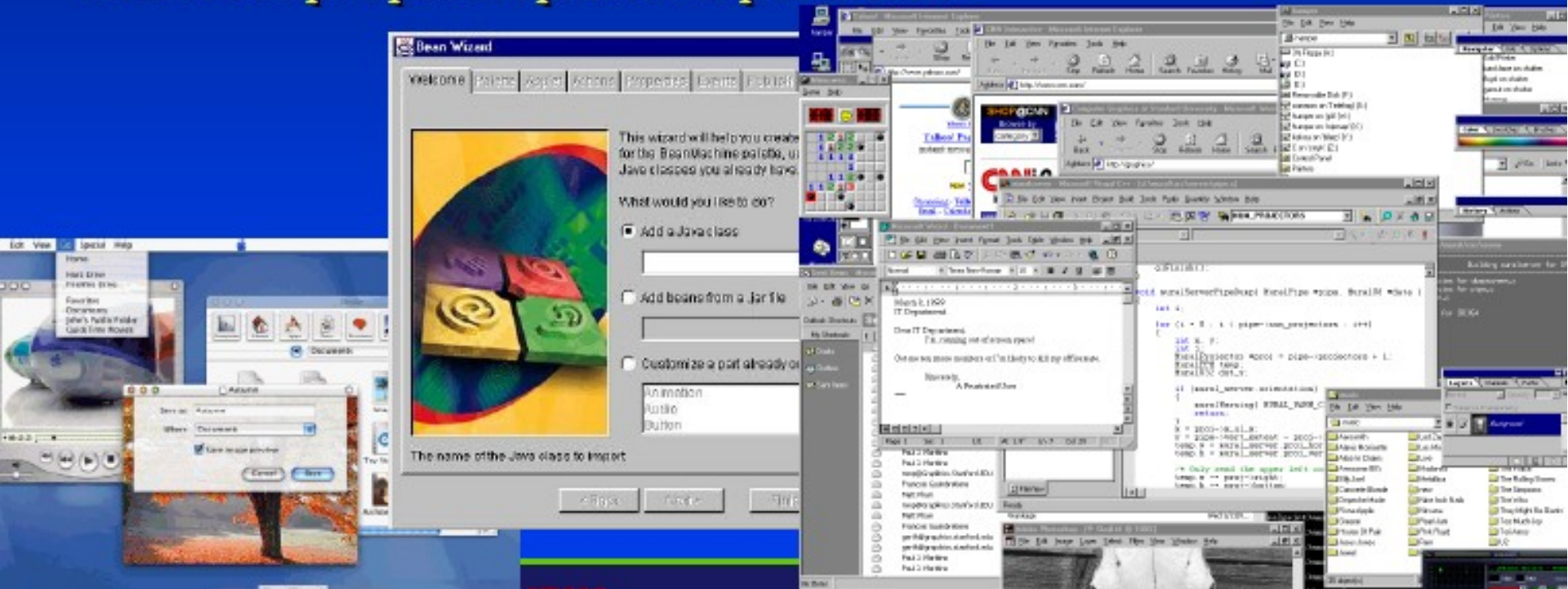
# Textile Industry

- Fashion design
- Real-time cloth animation
- Web-based virtual try-on applications



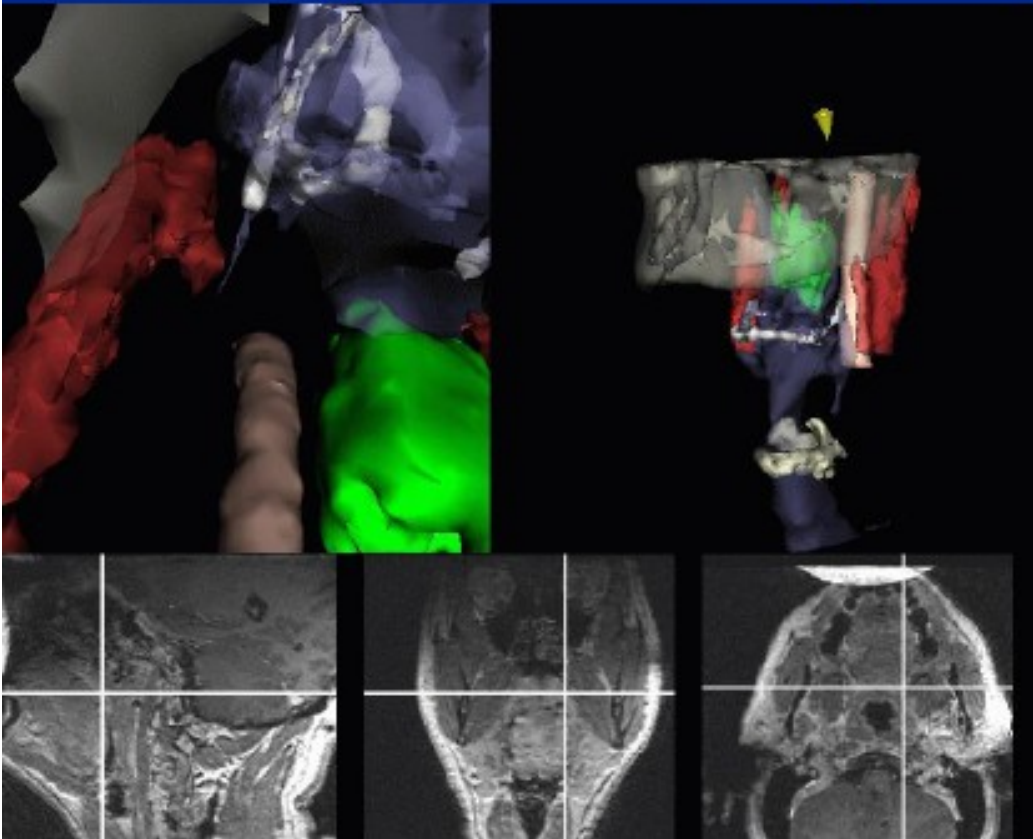
# Graphical User Interface: GUI

- Integral part of everyday computing
- Graphical elements everywhere
  - Windows, cursors, menus, icons, etc.
- Nearly all professional programmers must have an understanding of graphics in order to accept input and present output to users.



# Medical Applications

- Significant role in saving lives
- Training, education, diagnosis, treatment

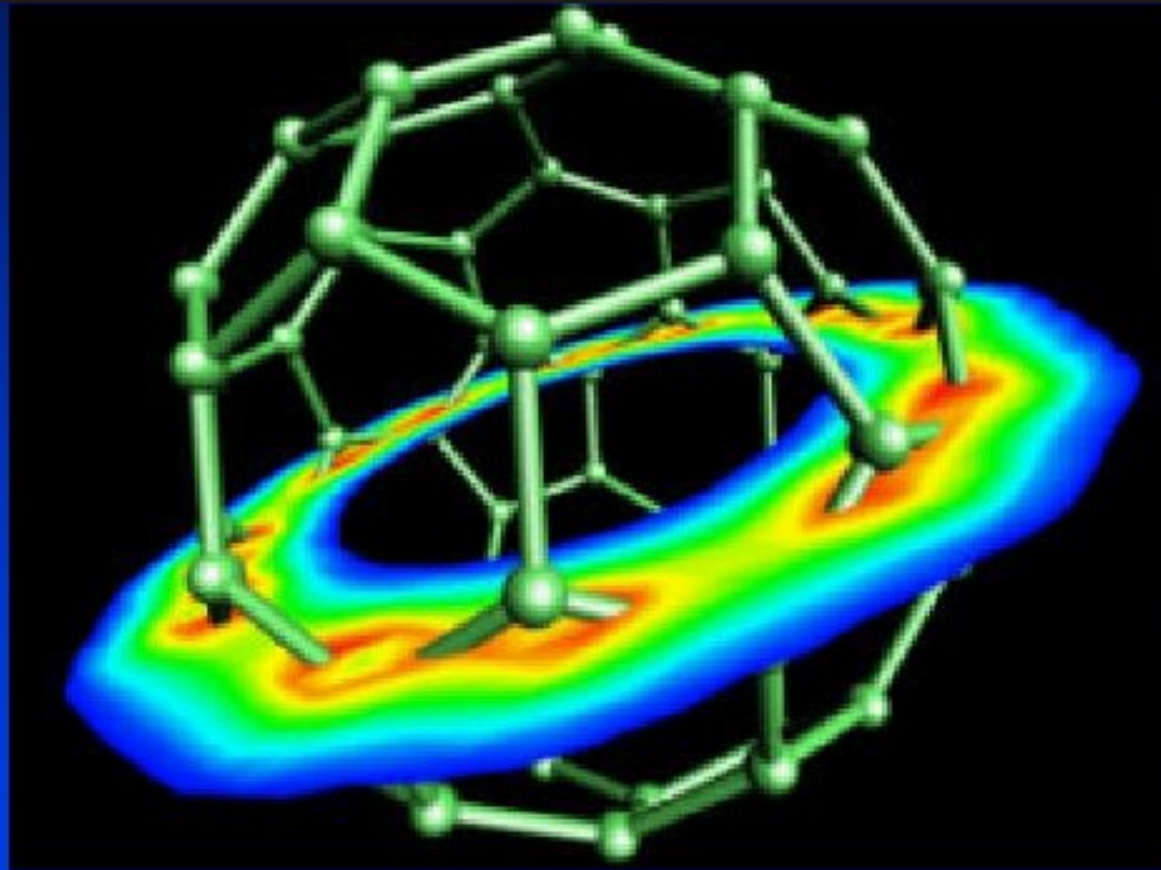


The Visible Human Project

Creation of complete, anatomically detailed 3D representation of human bodies.

# Scientific Visualization

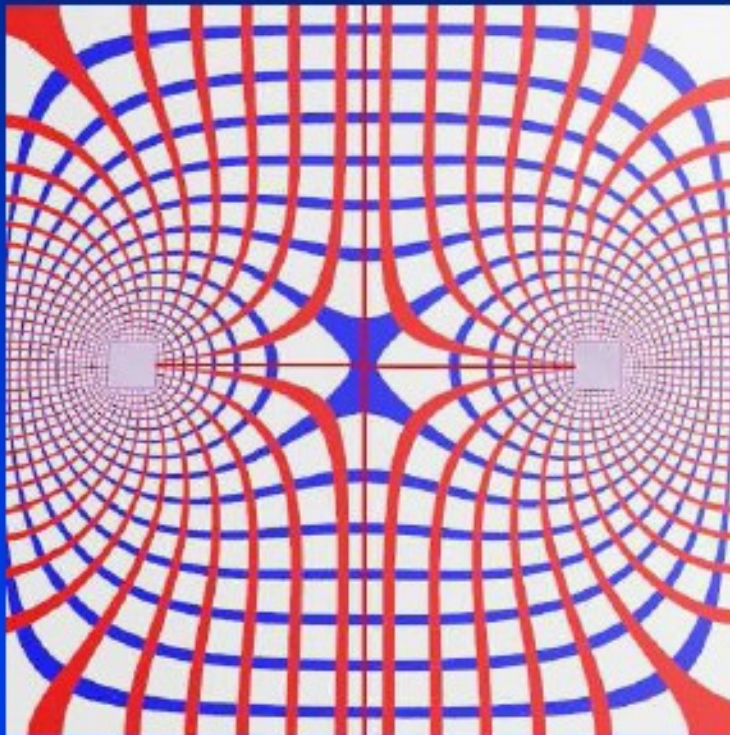
- Scientific data representation
- Picture vs. stream of numbers
- Techniques: contour plots, color coding, constant value surface rendering, custom shapes



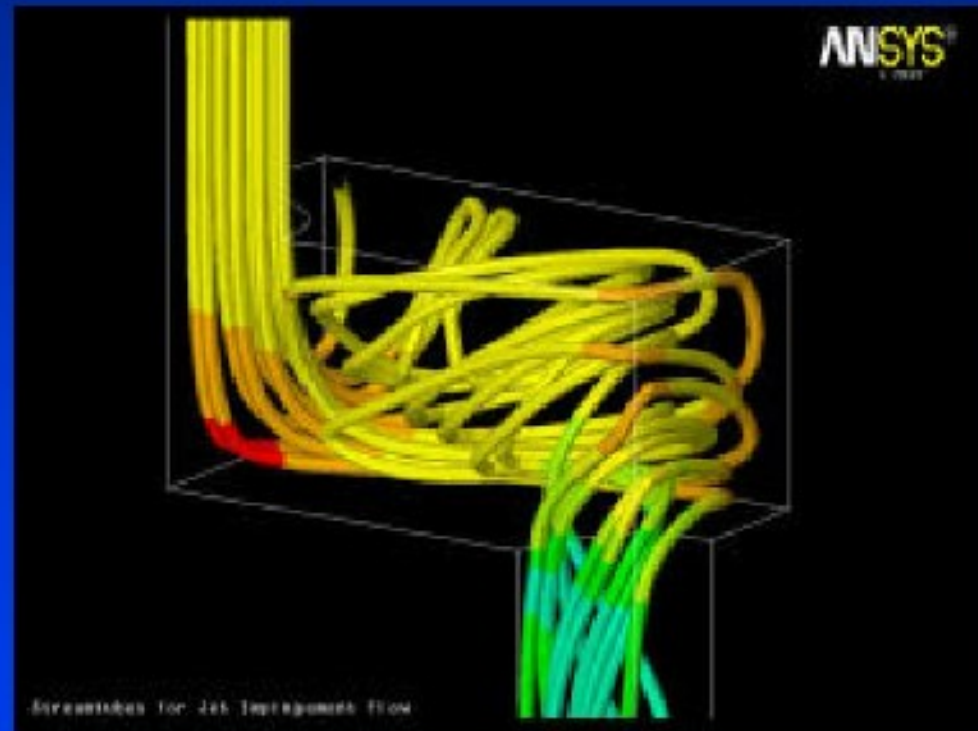
Display of a 2D slice through the total electron density of C-60; Created by Cary Sandvig of SGI

# Scientific Visualization / Simulation

Electromagnetic potential field



Computational Fluid Dynamics (CFD)



# Virtual Reality

- User interacts with objects in a 3D scene
- Special devices (input, output)
- Virtual walkthroughs
- Equipment training (pilots, surgeons, etc.)



Force reflecting gripper



Haptic devices



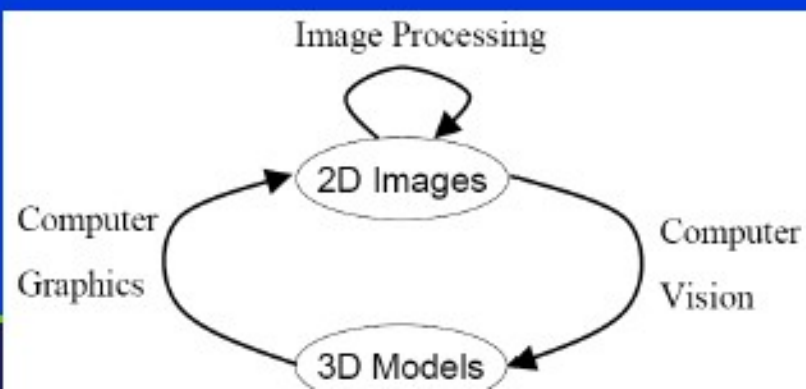
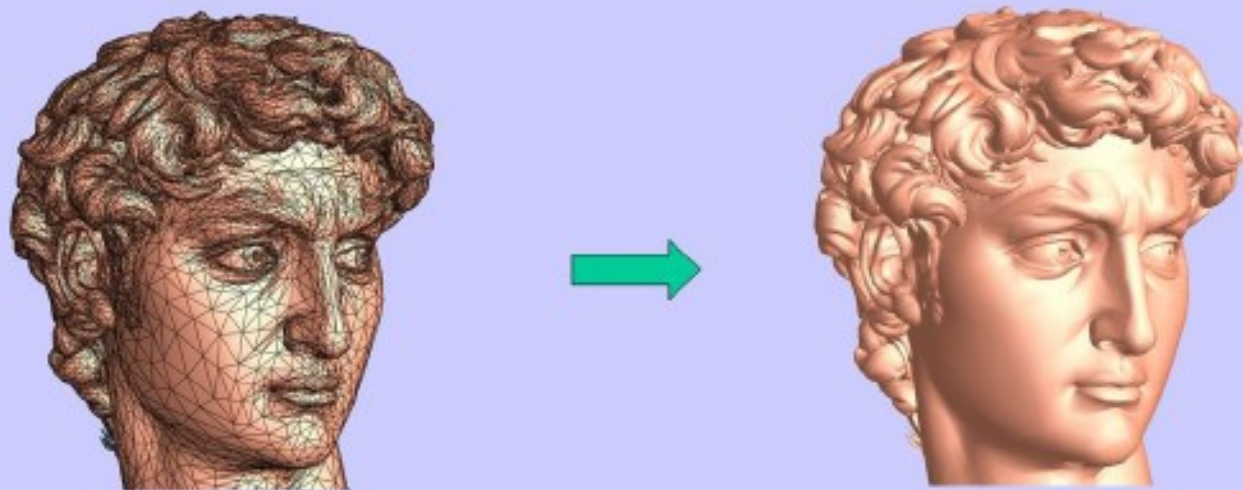
Force feedback exoskeleton



Haptic workstation



# Image Processing, Analysis, and Synthesis



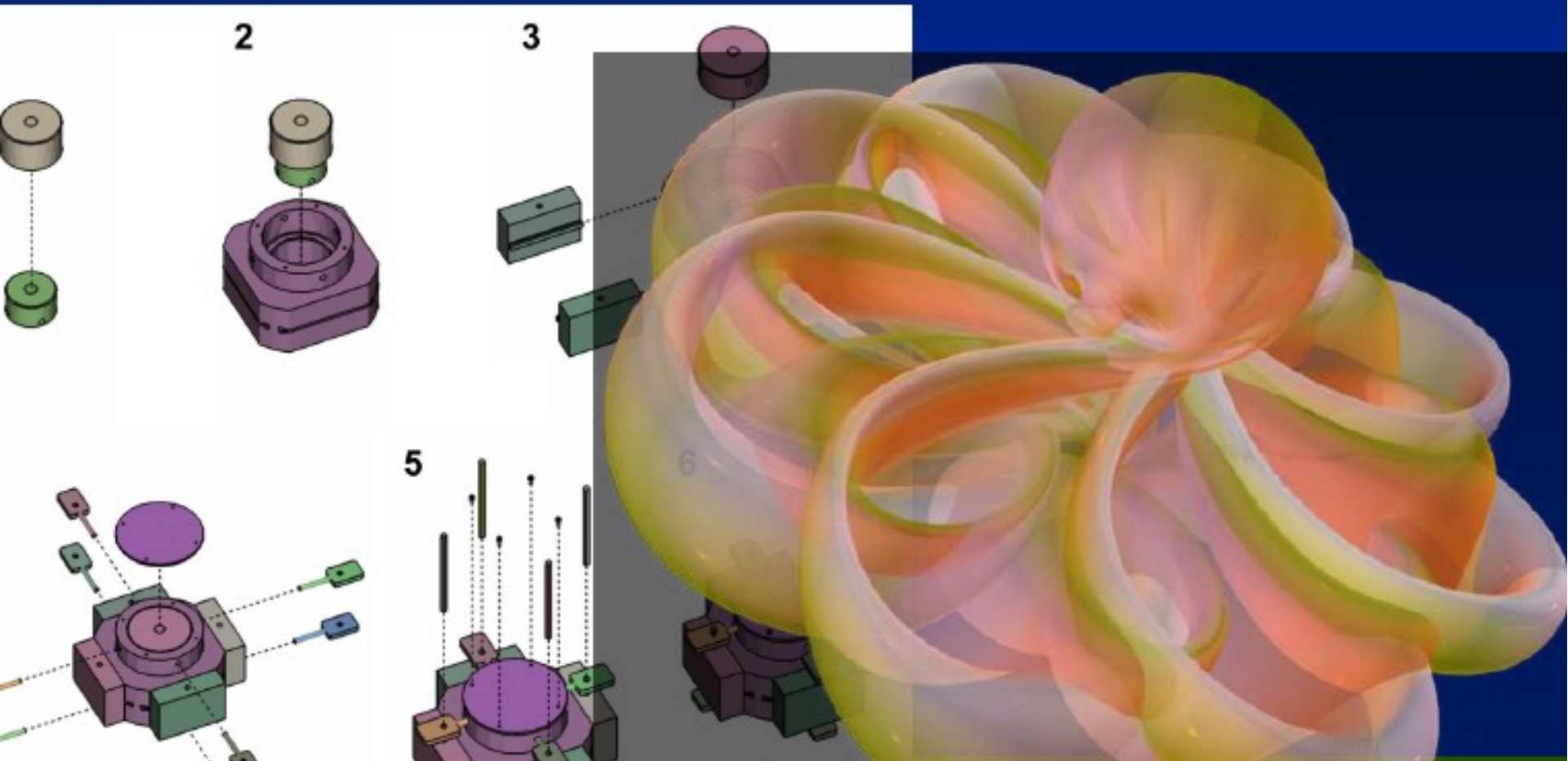
# Computer Art

- Digital Sculpting



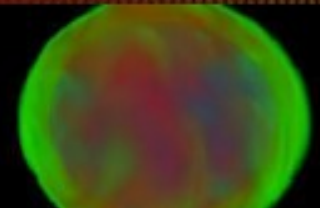
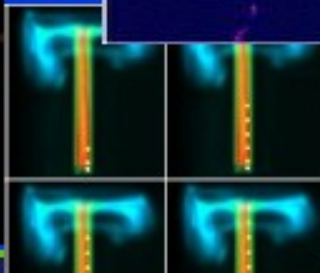
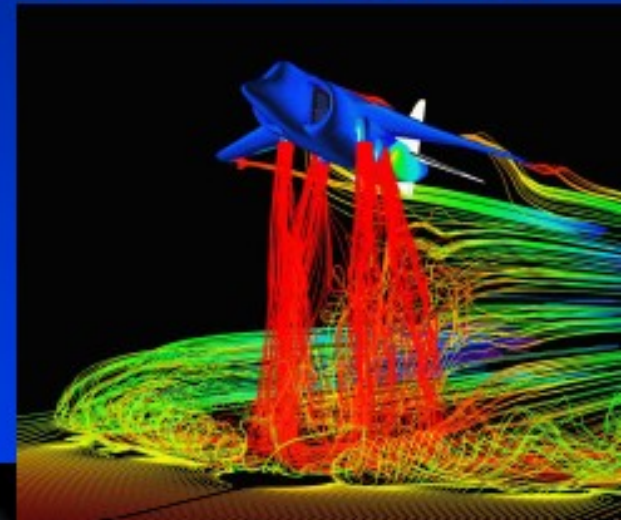
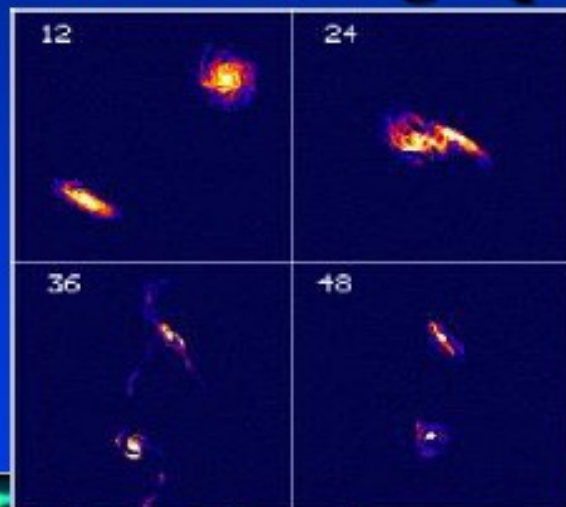
# Graphics Applications

- Training and education



# Why Visualization

**Visualization is a method of extracting meaningful information from complex or voluminous datasets through the use of interactive graphics and imaging**



# Why Visualization

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- Enable scientists (also engineers, physicians, general users) to observe their simulation and computation
- Enable them to describe, explore, and summarize their datasets (models) and gain insights
- Offer a method of SEEING the UNSEEN
- Reason about quantitative information
- Enrich the discovery process and facilitate new inventions

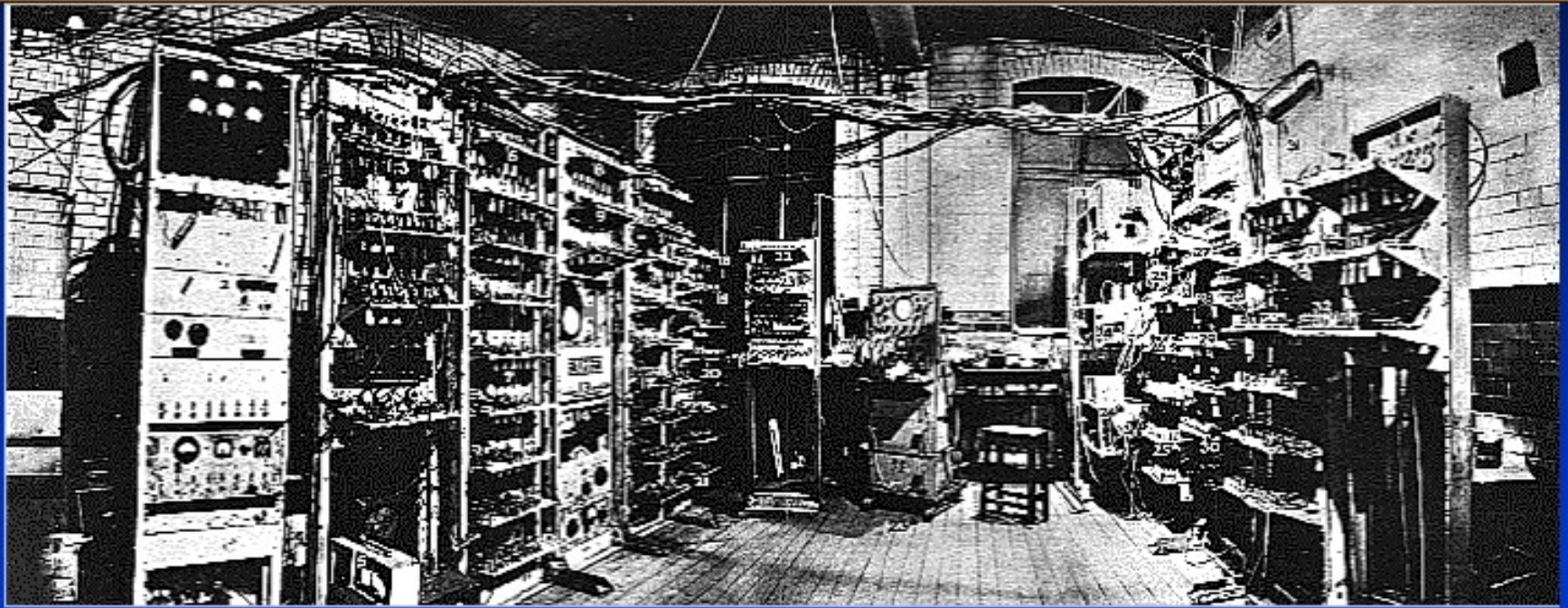
# History Computer Graphics

- Computer Graphics appeared with first digital computer in 40s which has CRT.
- However term was coined by William Fetter in 1960.
- Ben Laposky created first graphic image in 1950 using electronic machine.
- Vector display appeared in 1963.
- At the same time, Ivan Sutherland developed sketchpad, a drawing program for his PhD. thesis at MIT
- It uses light pen as input device and oscilloscope (vector scan) as output device.
- It was first interactive graphics system which can be used to input drawings and save in computer

# History Computer Graphics

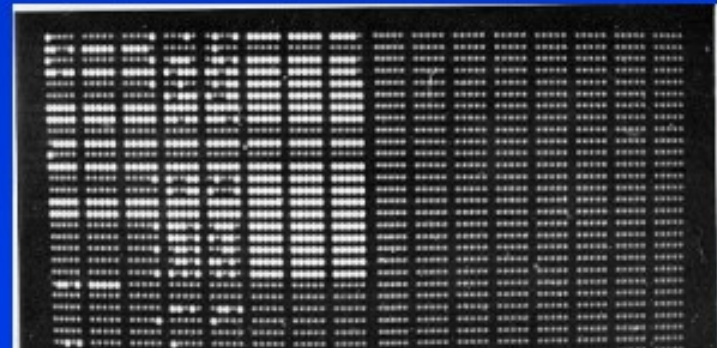
- SAGE air defence system used vector scanned monitor as its main output device and light pen as main input device.
- Vector display appeared in 1975 with the concept of frame buffer.
- With the invent of windows os in 1980s, graphics system improved drastically.
- Superpaint system was developed by Richard Shoup, Alvy Ray Smith.
- LCD appeared in 1990s.
- Then HMD appeared for VR.
- Nowadays we have MSpaint, Photoshop

# Where Are We Coming From: TEX



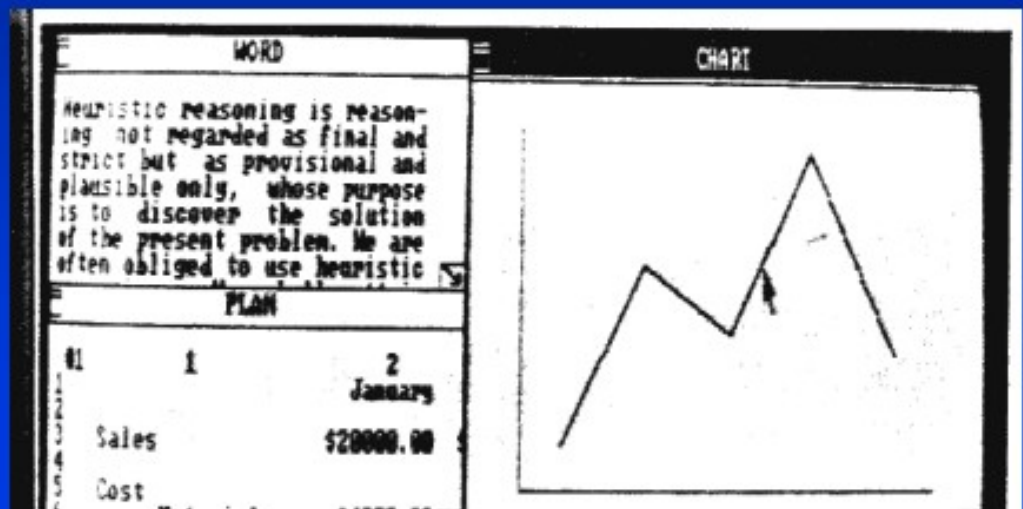
Manchester Mark I

Display



# From Text to GUIs

- Invented at PARC about 1975. Used in the Apple Macintosh, and now prevalent everywhere.



# Display Hardware

- Vector displays
  - 1963 – modified oscilloscope
  - 1974 – Evans and Sutherland Picture System



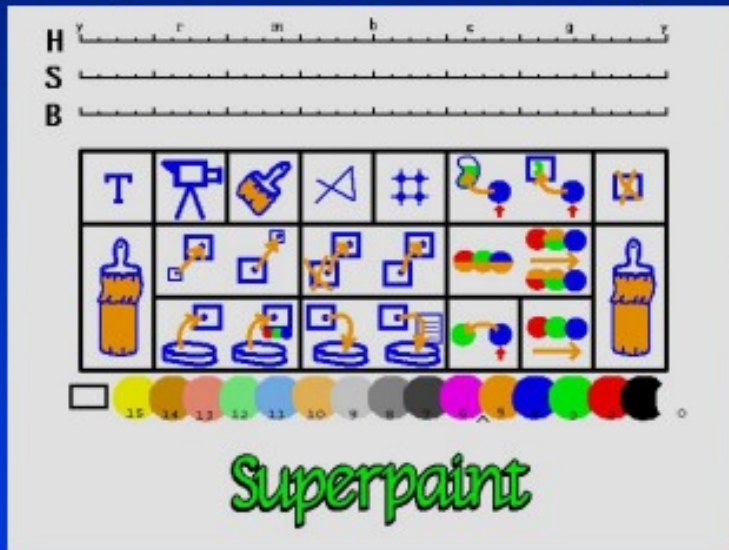
# Ivan Sutherland (1963) – SKETCHPAD Drawing

- Sketchpad (Sutherland, MIT 1963)
- First interactive graphics system
- Many of concepts for drawing in current systems
  - Pop-up menus
  - Constraint-based drawing
  - Hierarchical modeling



# Paint Systems

- SuperPaint system: Richard Shoup, Alvy Ray Smith (PARC, 1973-79)



- Nowadays, image processing programs like Photoshop can draw, paint, edit, etc.

# Image Processing

- Digitally alter images, crop, scale, composite
- Add or remove objects
- Sports broadcasts for TV (combine 2D and 3D processing)



# Display Hardware

- Raster displays
  - 1975 – Evans and Sutherland frame buffer
  - 1980s – cheap frame buffers → bit-mapped personal computers
  - 1990s – liquid-crystal displays → laptops
  - 2000s – micro-mirror projectors → digital cinema
- Others
  - stereo, head-mounted displays
  - autostereoscopic displays



# Graphics System

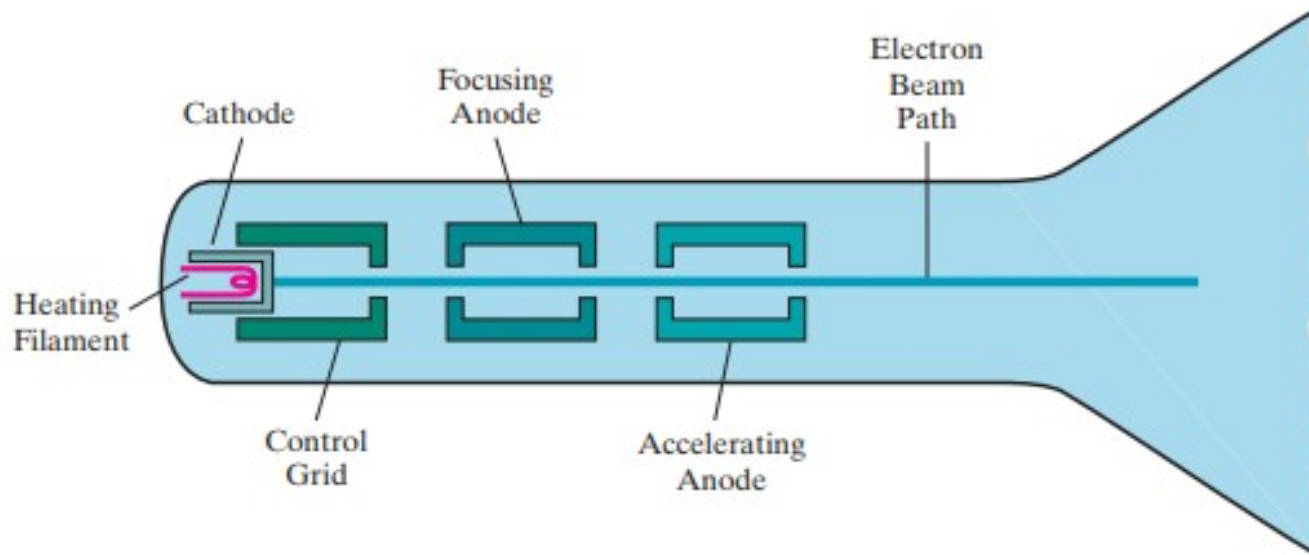
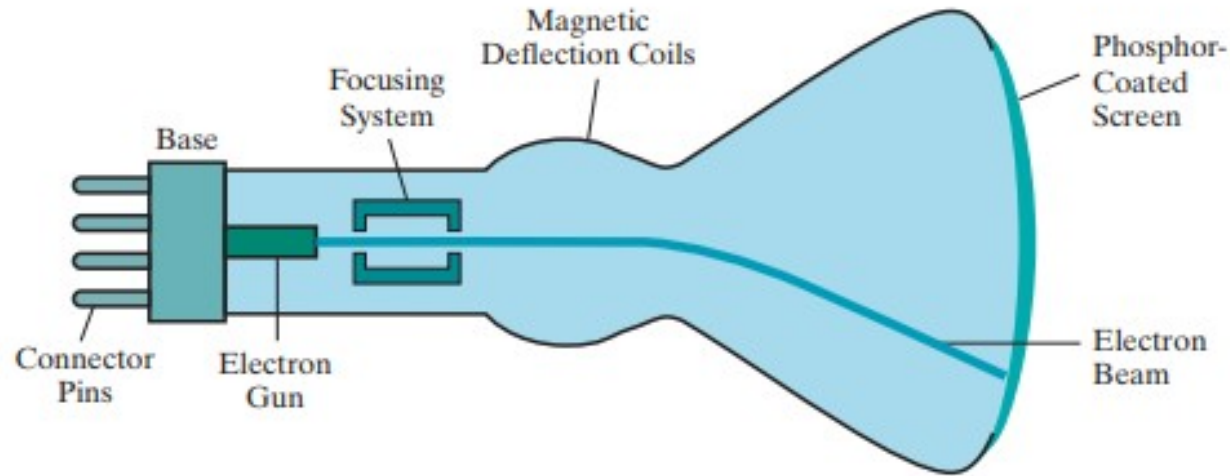
- Graphics system is responsible for synthesis and visualization of objects and data.
- The components of graphics system are
  - **Output device**
  - **Graphics Card**
  - **Input device**
  - **graphics software.**



# Output Device : CRT

- The primary output device is monitor screen.
- Monitors are softcopy output device.
- Others are projectors, printers, plotters, HMD (wireless)
- Monitors can be CRT, LCD, LED

# Output Device : CRT





# CRT

- Earliest technology for graphics output is CRT.
- The CRT consists of row of pixels known as scanning line.
- Each pixels consists of phosphorus.
- When phosphorus are struck by electrons to generate light of different intensities.
- These lights creates graphics or scene.
- CRT consists of heating filament, control grid, focusing anode and accelerating anode.



# CRT

- Control grid is negative charged that controls the generation of electrons.
- Focusing anode is positively charged that is used to focus the scatter electrons.
- Accelerating anode is positively charge that is used to accelerate the electronic beam.
- Heating filament heats cathode and generates the scattered electrons



# Graphics Card

- Graphics card consists of important components such as display controller, video memory, video controller

# Display Controller

- Display controller is also known as display processor or graphics processor.
- It takes input from host controller and various input devices.
- Input devices helps to read the coordinates of the the object.
- So, objects or scene is represented with points, lines, arcs.
- Then it performs multi-stage operations to synthesize image.
- It can perform different transformations operations to render the scene in desired form.



# Display Controller

- This process goes on till the computer is on.
- That is why dedicated display controller is necessary.
- CPU assigns any graphics rendering task to display controller.
- Display controller takes all load from CPU.



# Video Memory

- For graphics processing system memory can be shared between GPU and display controller or can have dedicated video memory.
- Frame buffer and depth buffer together is known as video memory.
- The image generated by display controller is stored in frame buffer.
- In fact frame buffer consists of color information of the image.
- Some part of video memory stores code for graphics operation and data.



# Video Controller

- It converts digital image into analog voltage.
- Screen needs analog voltage to display images.
- Voltage is generated according to 0's and 1's stored in frame memory.
- It means it activates suitable electro-mechanical mechanism so that pixels on screen emit light.
- Excitation of the pixels on the screen depends on the value in frame buffer.



# Types Graphics System

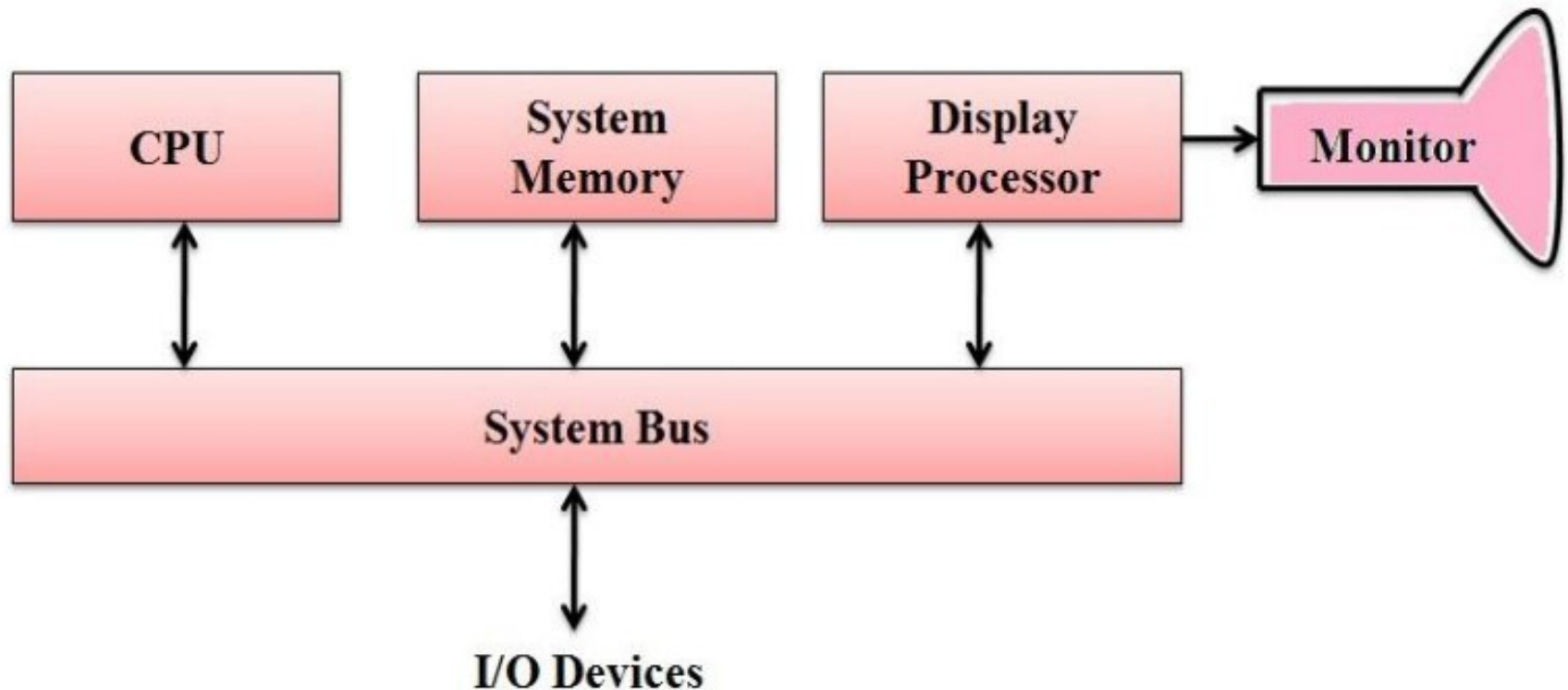
- Based on the method used for generation of the scene, graphics system can be
- Vector Scan Graphics
- Raster Scan Graphics



# Vector Scan

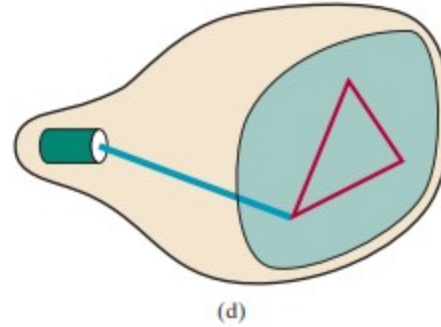
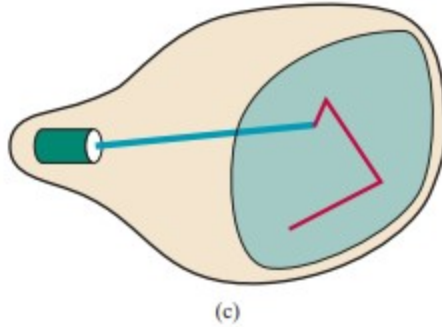
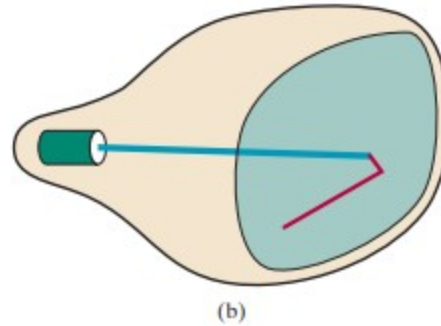
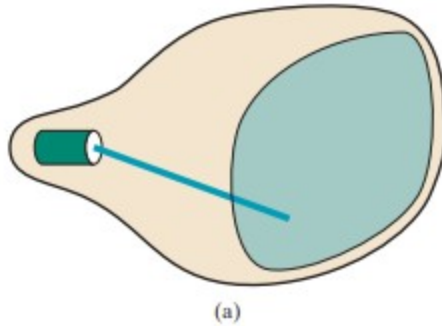
- In vector scan display image is represented using continuous geometric primitives such as lines and curves.
- Image is rendered by rendering these primitives.
- Such images are known as vector image.
- A vector scan device excites only those pixels of the screen that are part of these primitives.

# Random-Scan Systems/ Random-Scan Display Processor



**Figure 11: Architecture of a simple random scan system**

# Vector Scan





# Vector Scan: Pros and Cons

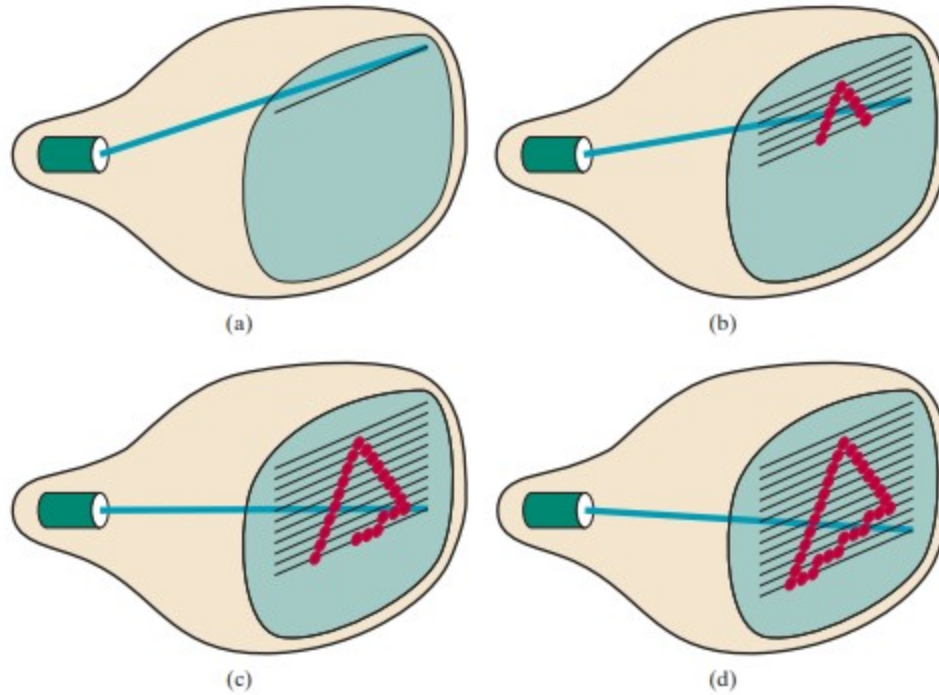
- Excites selected pixels only.
- This requires high precision and complex hardware.
- So, costly.
- For complex scenes, flicker is visible as rendering involves lots of random movement.
- Good for wireframe (outline) images.



# Raster Scan

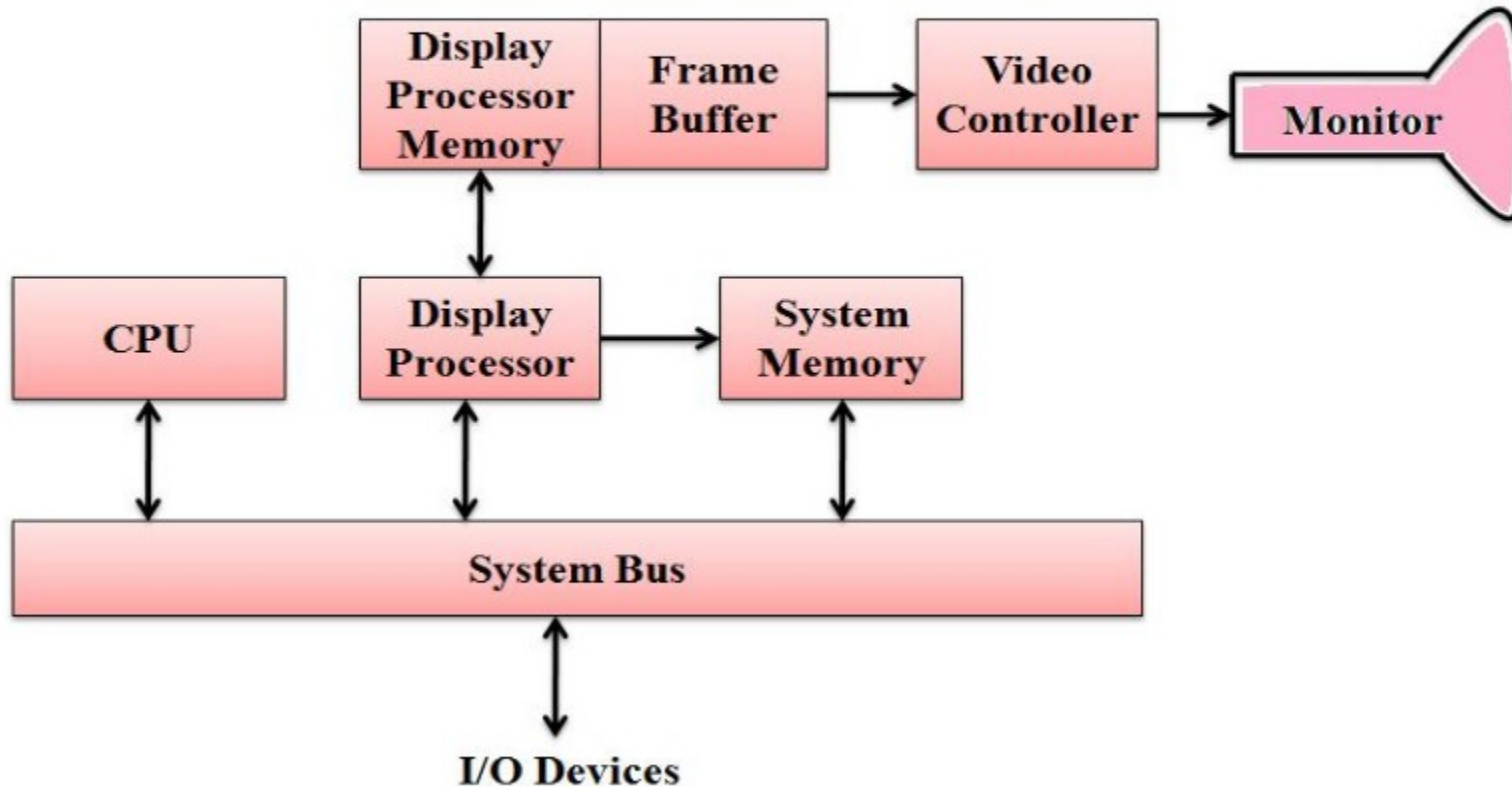
- In raster scan, image is represented as the grid of pixels.
- Such image is known as raster image.
- To render raster image, all the screen pixels are scanned from left to right, top to bottom.
- Some pixels are excited and some are left according to image.
- Each row in the pixel grid is called scanning line.

# Raster Scan



# Raster Scan Display Processor

- Figure 9 shows one way to set up the organization of a raster system containing a separate **display processor**, sometimes referred to as a **graphics controller** or a **display coprocessor**.



**Figure 9: Architecture of a raster-graphics system with a display processor**



# Raster Scan : Frame Buffer

- Frame buffer stores image generated by display processor.
- Contains pixels equal to the screen resolution.
- Display processor works at the speed of CPU (nanosecond).
- Video controller is much slower (millisecond)
- Mismatch in speed.
- Image may distorted if directly output.
- To synchronize operation, multiple frame buffers are used.

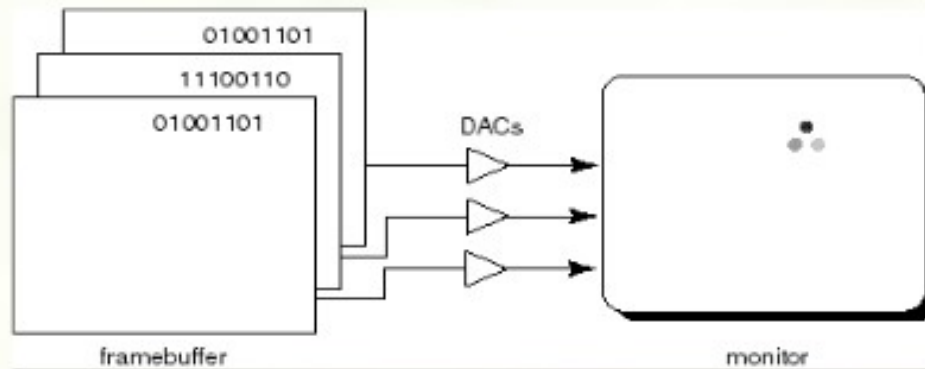


# Raster Scan : Frame Buffer

- At least two buffers used.
- Primary buffer and secondary buffer.
- Initially, display controller fills secondary buffer.
- Then primary buffer is filled.
- Video controller takes input from primary buffer.
- Then role reversed.



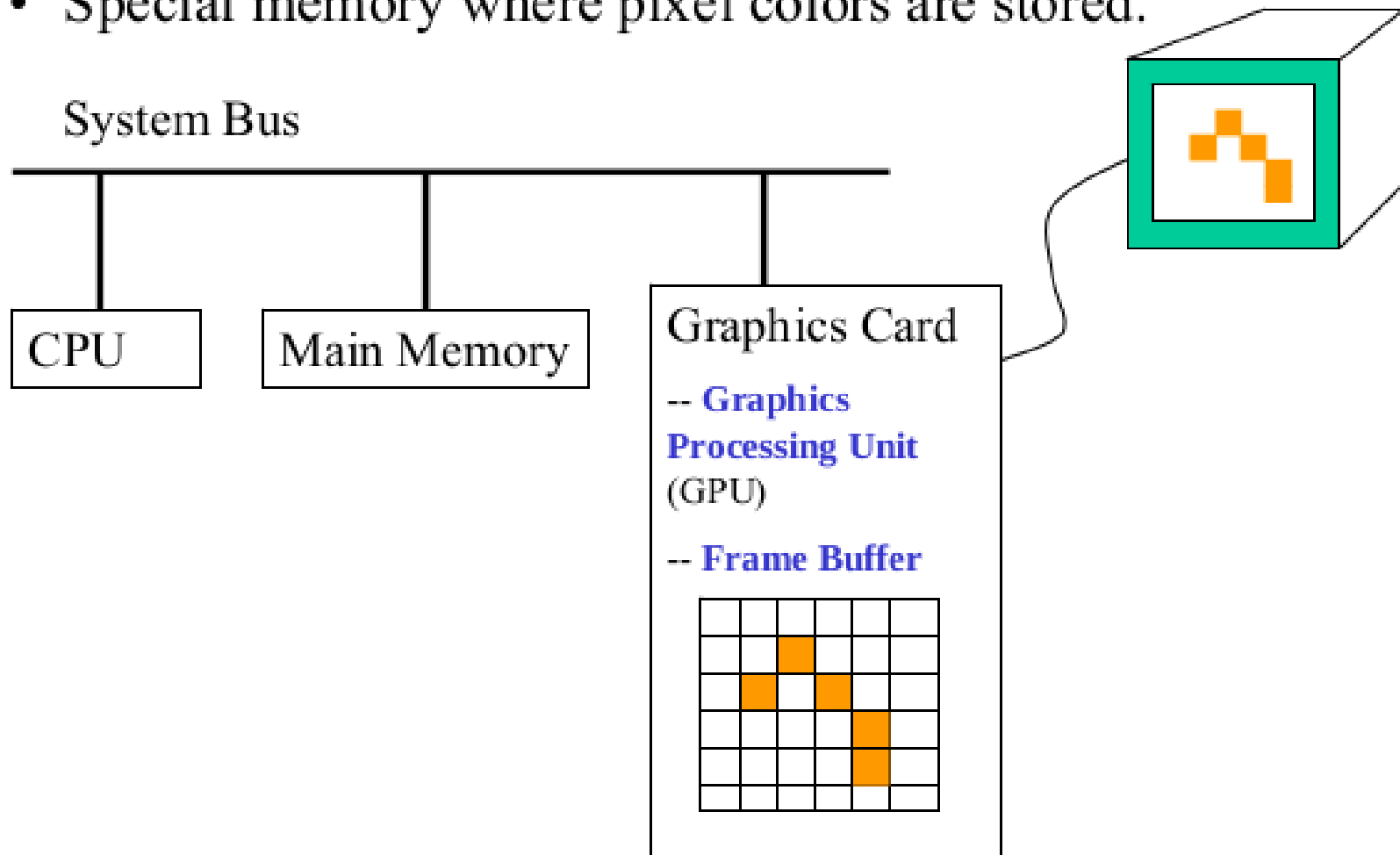
# RGB framebuffer



- The term **true-color** is sometimes used to refer to systems which the framebuffer directly stores the values of each channel.
- As memory prices have fallen, true-color has become standard.

| Age Group | Percentage |
|-----------|------------|
| 18-24     | 18%        |
| 25-34     | 22%        |
| 35-44     | 15%        |
| 45-54     | 12%        |
| 55-64     | 10%        |
| 65-74     | 8%         |
| 75-84     | 5%         |
| 85+       | 3%         |

- Special memory where pixel colors are stored.





# Screen Refreshing

- Light due to pixel excitement starts decaying over time.
- This lead to fading of the scene.
- Moreover, pixels get excited at different point of time.
- This lead to image distortion.
- So pixels are kept excited periodically called refreshing.
- Typical refresh rate is 60hz.



# Raster Scan : Pros and Cons

- Simple hardware to implement.
- Good for generating complex images.



# Graphics Input

- Allows to input data and interact.
- Eg. ubiquitous keyboard and mouse.
- Speech interaction (speech recognition and synthesis)
- Haptic interaction (smart phones)



# Graphics Input

- **Keyboard**
- Primarily used for entering text
- Inputs non graphic data.
- It can enter screen coordinates, menu selection, graphics functions
- Cursor keys can be used to select displayed objects.



# Graphics Input

- **Mouse**
- Moves screen cursor
- Records amount and direction of movement

# Graphics Input

- ✎ **Tablet:** locate input primitives. A **tablet** provides an area on which the user can slide a stylus. The tip of the stylus contains a micro switch. By pressing down on the stylus the user can trigger the locate.



# Graphics Input

## Touchscreen

- A **touchscreen** is an electronic visual display that can detect the presence and location of a touch within the display area.
- The term generally refers to touching the display of the device with a finger or hand.



# Graphics Input

## Digital Camera

- A camera that stores the pictures or video it takes in electronic format instead of to film.
- Digital cameras have become the camera solution for most users today as the quality of the picture they take has greatly improved and as the price has decreased.



# Graphics Input

## Light Pen

- A **light pen** is a computer input device in the form of a light-sensitive wand used in conjunction with a computer's CRT display.
- It allows the user to point to displayed objects or draw on the screen in a similar way to a touchscreen but with greater positional accuracy.





# Graphics Input

- **Light Pen:**
- detects the light coming from screen and selects the screen position.
- Sensitive to short burst of light coming from phosphorus coating.
- Other light sources are not detected.
- Arm fatigue.
- Screen is obscured by and and pen
- Cannot detect positions within black area.

# Graphics Input

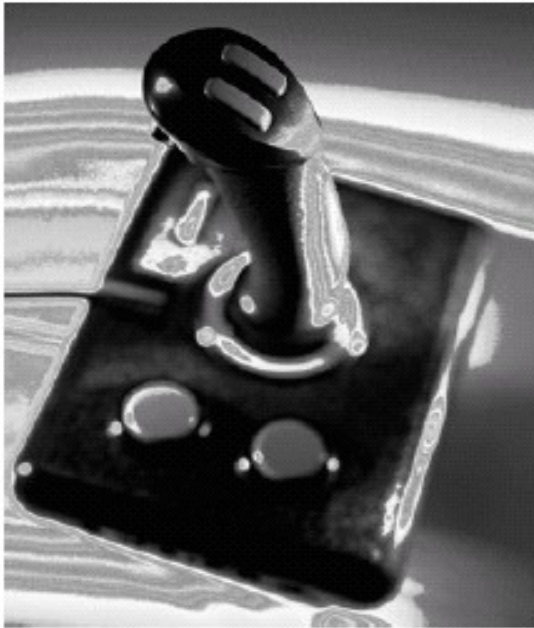
## Barcode Reader

- A **barcode reader** or **scanner** is a hardware device capable of reading a barcode and printing out the details of the product or logging that product into a database.



# Graphics Input

- **Joystick and Trackball:** locate and valuator devices.





# Graphics Input

- **Joystick:**
  - Steer screen cursor.
  - Select screen position with stick movement.
- **Track Ball:**
  - Ball is rotated to produce cursor movement.

# Data Glove

- Constructed with a series of sensors that can detect hand and finger motions.
- The transmitting and receiving antennas can be structured as a set of three mutually perpendicular coils, forming a three dimensional Cartesian coordinates system.
- Electromagnetic coupling between the three pairs of coil is used to provide information about the position and orientation of hand.





# Graphics Software

- Graphics library
- Application programs



# Graphics library

- High level programming language usable graphics library provides function to work with graphics.
- These function can be used to generate picture components such as straight lines, polygons, circles.
- Some function can be used to set color, intensity values and applying transformation.



# Graphics Application Software

- Application programs to manipulate graphics without programming.
- Paint, CAD are some examples.



# Graphics Standard

- Standardized software determines its portability, reliable, and efficiency.
- It means can be executed in most of hardware system.
- It also maintains quality.
- It also maintains consistency in all graphics applications.
- It is result of consensus among large community that takes many years.



# GKS

- GKS (Graphical Kernel System):
- First standard by international standard organization introduced in 1984.
- GKS provides a set of functions for the purpose of generating 2D pictures on vector or raster devices.
- It also supports graphical input and its modification.
- It provides interfaces between an application program and a graphical input output devices.
- Provides interactive and non-interactive graphics functions.



# GKS

- The application programmer constructs application programs that uses GKS in order to complete some task or problem.
- Application user uses the graphics application to get problem done.



# PHIGS

- Programmers' Hierarchical Interactive Graphics Standard.
- Proposed by ANSI in 1985.
- Offers commands for modeling as well as for drawing.
- Color specifications and surface rendering.
- A 3D extension of GKS
- Increased capabilities for object modeling, color filling, surface rendering.
- It has concept of logical and physical devices.
- Which frees the need of software updation whenever new device is added.



# OpenGL

- Open Graphics Library.
- Can generate and manipulate 2D and 3D graphical objects.
- Fast and portable.
- OpenGL commands quickly and easily construct and manipulate graphics objects and scenes.
- Consists of over 250 functions.
- Developed by Silicon Inc. in 1992.